Lecture 29 (*May 3*)

Reading: Section 12.1
Viewing: none

We will now start discussing major applications of linear algebra: probability and statistics. First of these is **probability**, which studies the likelihood of future events. Suppose we're in a situation with n possible outcomes $x_1, \ldots, x_n$ (think of these as real numbers), which arise with probabilities $p_1, \ldots, p_n$, respectively. Then:

$$p_1, \ldots, p_n > 0 \qquad \text{and} \qquad p_1 + \cdots + p_n = 1 \tag{266}$$

Definition 24. *The **mean** is the sum of the possible outcomes, weighted by their probabilities:*

$$\mu = p_1 x_1 + \cdots + p_n x_n \tag{267}$$

*and the **variance** is:*

$$\Sigma = p_1(x_1 - \mu)^2 + \cdots + p_n(x_n - \mu)^2 \tag{268}$$

*The square root of the variance, namely $\sigma = \sqrt{\Sigma}$, is called the **standard deviation**.*

The mean is sometimes called the **expected value**, which is the plain English significance of this notion. The variance is a positive quantity which measures how far away from the mean our outcomes are. Clearly, the variance is 0 only if there exists a single outcome with probability 1.

The situation described above is when there are finitely many (or discrete) outcomes. But we could also study a quantity which takes infinitely many (or continuous) values, for example temperature. In this case, the probability is encoded not in a finite set of numbers, but in a function:

$$p(x) : \mathbb{R} \longrightarrow \mathbb{R}_{\geq 0}$$

called **probability distribution**. In this context, you shouldn't measure the probability of your quantity being exactly equal to x (for example, it makes little sense to measure the probability that temperature will be exactly 62.746 degrees) but you measure instead the probability that the quantity lies in a certain interval $[a, b]$ (for example, you measure the probability that temperature falls between 62 and 63 degrees):

$$\text{Prob}(x \in [a, b]) = \int_a^b p(x)dx \tag{269}$$

The reason for the integral is that it is the continuous analogue of a sum: if you subdivide the interval $[a, b]$ into many (say N very large) small intervals, then the integral in (269) is infinitesimally close to the sum of the probabilities that x lies in the N small intervals. In this continuous context, condition (266) is replaced by:

$$\int_{-\infty}^{\infty} p(x)dx = 1 \tag{270}$$

while the mean (267) and the variance (268) must be defined by integrals:

$$\mu = \int_{-\infty}^{\infty} p(x)x dx \tag{271}$$

$$\Sigma = \int_{-\infty}^{\infty} p(x)(x-\mu)^2 dx \tag{272}$$

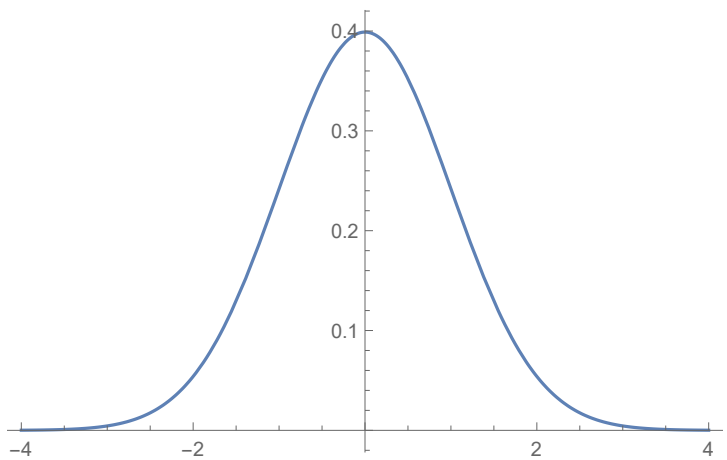
The most basic example of probability distribution is the uniform one, where the variable can lie in an interval (say $[0, c]$) with constant probability:

$$p(x) = \begin{cases} \frac{1}{c} & \text{if } x \in [0; c] \\ 0 & \text{otherwise} \end{cases}$$

However, this probability distribution does not really show up in nature. Real-life probabilities do not like the sharp cut-offs that the function above experiences at the endpoints of the interval $[0, c]$. Perhaps even more importantly, real-life probability distributions tend to be higher around the mean and lower away from the mean. The standard example of such a behavior is the **normal (or Gaussian) distribution**:

$$p(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \tag{273}$$

whose graph (sometimes called a "bell curve") is plotted below. As always, the probability that the variable x lies in a given interval $[a, b]$ is equal to the integral of $p(x)$ (or the area under the graph) from $x = a$ to $x = b$.



The reason for dividing by the factor $\sqrt{2\pi}$ is because you want the total probability to be equal to 1, i.e. (270). The mean and variance of the normal distribution (273) are given by:

$$\mu = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-\frac{x^2}{2}} dx = 0$$

$$\Sigma = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx = 1$$

More generally, the normal probability distribution with arbitrary mean $\mu \in \mathbb{R}$ and arbitrary variance $\Sigma > 0$ is given by the function:

$$p(x) = \frac{e^{-\frac{(x-\mu)^2}{2\Sigma}}}{\sqrt{2\pi\Sigma}} \tag{274}$$

Normal distributions are extremely important, because of the **central limit theorem**: if you perform an experiment $N \to \infty$ times, the mean of your samples tends to look as if it came from a normal distribution. For example, if you flip a coin N times, the probability of getting k heads is close to the probability that $x \in [k, k+1]$ for the normal distribution (274) with mean $\mu = \frac{N}{2}$ and variance $\Sigma = \frac{N}{4}$. In the error analysis of the probability distribution (274), the probability that:

$$\begin{aligned}
&x \text{ is at most } \ \sigma \text{ away from the mean is } \sim 68\% \\
&x \text{ is at most } 2\sigma \text{ away from the mean is } \sim 95\% \\
&x \text{ is at most } 3\sigma \text{ away from the mean is } \sim 99.7\% \\
&x \text{ is at most } 4\sigma \text{ away from the mean is } \sim 99.99\% \\
&x \text{ is at most } 5\sigma \text{ away from the mean is } \sim 99.9999\%
\end{aligned}$$

You can actually see this in the graph of the bell curve above: about 99.7% of the area under the graph of the bell curve is between $x = -3$ and $x = 3$ etc.